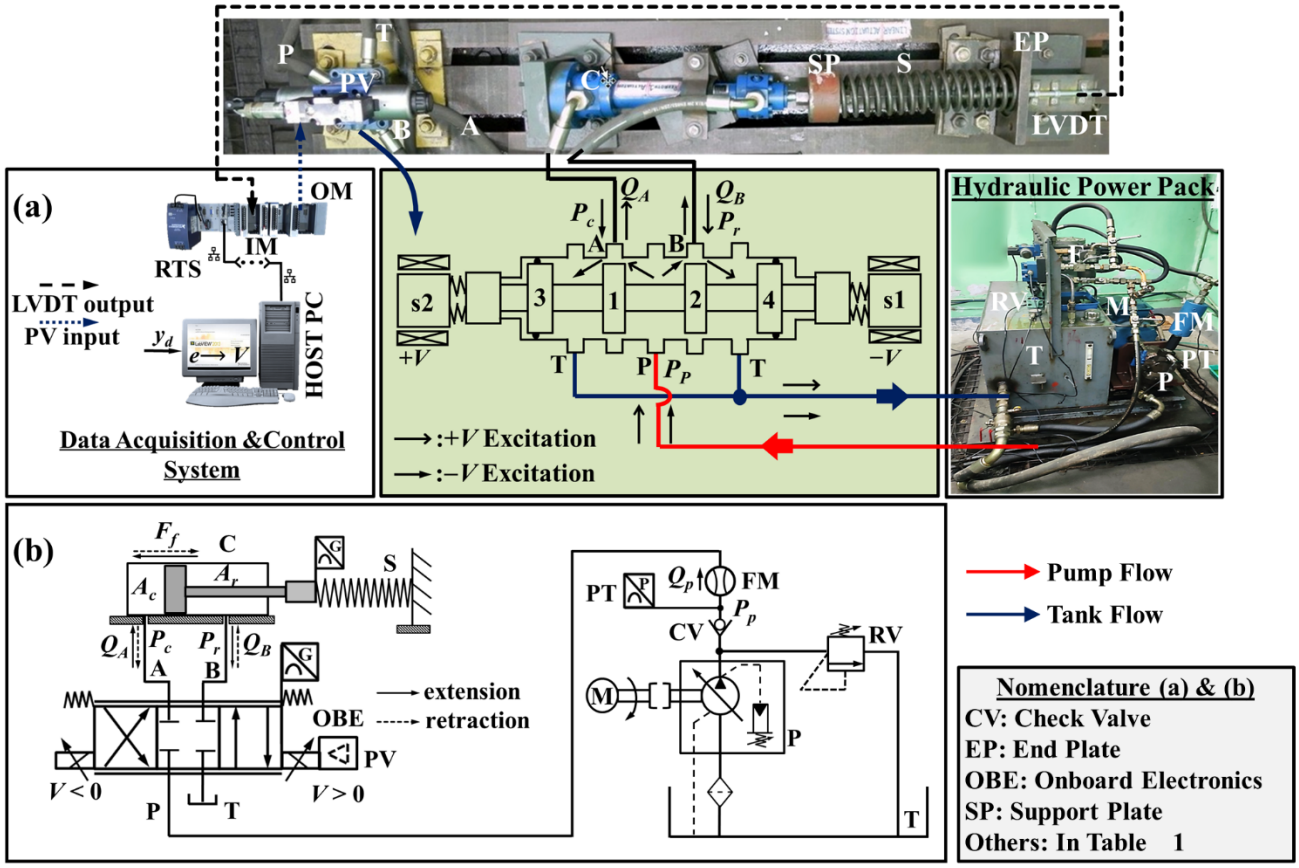


Figure 1: Experimental setup for electrohydraulically actuated linear-motion study in (a) photographic view and (b) circuit diagram

Table 1 Component specifications of Single-Cylinder Test Bench

Component	Make	Specifications
Proportional valve PV	Rexroth	4WREE 10 E50-23/G24K31/A1V
Single-rod cylinder C	Rexroth	Bore 40mm and rod 20mm in diameter, piston mass 2.5kg
Variable-displacement pump P	Rexroth	A10VSO 45DR/31; $4.5\times10^{-5}m^3$ nominal displacement; maximum discharge $8.3\times10^{-4}m^3s^{-1}$
Flowmeter FM	Kral	Working pr.0–2.5×10^7 Pa; flow 2–$1.7\times10^{-3}m^3s^{-1}$; maximum temp 120 °C; accuracy ±0.1%
Input Module IM	NI Systems	NI 9215 (4 Channel; ±10 V A/D, 16-bit simultaneous analog input)

Linear variable differential transformer LVDT	HBM	Plunger type; 0-0.200m range with 0.5-10V analog output; Nominal Sensitivity: 80mV/V
Motor M	ABB	3GBA201044-ADEIN; 3-phase; 2 pole: 37kW; 2945rpm
Oil tank T	Local	Dim.-1.0m x 0.6m x 0.5m; capacity-200L
Output module OM	NI Systems	NI 9263 (±10V D/A,16-bit simultaneous analog output)
Pressure Relief valve RV	Yuken	EBG-03-H-11; 2.05×10^7 Pa
Pressure Transducer PT	Kistler	0–2×10^7 Pa; temp: -20 to 85°C; linearity, hysteresis, and repeatability ±0.25 %
Real-time system RTS	NI Systems	1GHz, 32-bit NI-cRIO 9081, Sampling rate-100 KS/s
Spring S	Local	Stiffness 20 kNm^{-1}

If the total moving mass of the assembly is m, then the consequent dynamics of the piston can be expressed as

$$\dot{y} = v, \tag{1a}$$

$$\text{and } \dot{v} = \ddot{y} = (P_c A_c - P_r A_r - ky - F_f)/m, \tag{1b}$$

$$\text{where } A_c = \pi d_c^2/4 \text{ and } A_r = \pi(d_c^2 - d_r^2)/4. \tag{1c}$$

In the above equations, y, $\dot{y} = v$ and $\ddot{y} = \dot{v}$ are the displacement, velocity, and acceleration of the piston respectively and d_c and d_r are the cylinder bore diameter and piston rod diameter respectively.

A friction model can be represented by a simpler version

$$F_f = \{F_c\}\,\mathrm{sgn}(\dot{y}) + \alpha_v \dot{y}, \tag{1d}$$

Using the orifice flow leakage model the overall flow-pressure model for the metered ports is hence expressed as

$$Q_A = A_c \dot{y} = c_{vA} V \sqrt{P_P - P_c} - Q_{lA} \text{ for } \dot{y} \geq 0, \tag{2a}$$

$$\text{or, } Q_A = A_c \dot{y} = c_{vA} V \sqrt{P_c} - Q_{lA} \text{ for } \dot{y} < 0, \tag{2b}$$

$$Q_B = A_r \dot{y} = c_{vB} V \sqrt{P_r} - Q_{lB} \text{ for } \dot{y} \geq 0, \tag{2c}$$

or, $Q_B = A_r\dot{y} = c_{vB}V\sqrt{P_P - P_r} - Q_{lB}$ for $\dot{y} < 0$. (2d)

where P_P is the pressure at the pump port **P** of **PV,** P_c and P_r are the cap and rod end pressure respectively and further considering P_T to be zero. The leakage through the valve is expressed for the orifice model as

$Q_{lA} = c_{lA}\sqrt{P_c}$ for $\dot{y} \geq 0$ and $c_{lA}\sqrt{P_p - P_c}$ for $\dot{y} < 0$, (2e)

and $Q_{lB} = c_{lB}\sqrt{P_p - P_r}$ for $\dot{y} \geq 0$ and $c_{lB}\sqrt{P_r}$ for $\dot{y} < 0$. (2f)

List of Parametric values:

$m = 2.5\text{kg}$

$d_c = 0.04\text{m}$

$d_r = 0.02\text{m}$

$F_c = 51.1\text{N}$

$\alpha_v = 2000\ \text{Ns/m}$

$c_{vA} = 3.46 \times 10^{-8}$

$c_{vB} = 1.73 \times 10^{-8}$

$c_{lA} = 3.8 \times 10^{-9}$

$c_{lB} = 1.9 \times 10^{-9}$

$P_p = 120\text{bar}$

$k = 20000\text{N}$

The initial condition for cap end and rod end pressure should be kept half of fixed pump pressure P_p, and its upper limit should not cross P_p and lower limit should not be less than zero. The voltage V limit is within $\pm 10\text{V}$